

Muhammad Hassan Askari

Computer Science Graduate | AI/ML & Trustworthy AI

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PROFILE

Computer Science graduate from COMSATS University Islamabad with a CGPA of 3.65/4.00 and research experience in machine learning, computer vision, and robust AI systems. Undergraduate research focused on traffic sign classification under snow-degraded conditions, with experience in deep learning, experimentation, model evaluation, and scientific writing. More recently, developed an LLM-based RAG application and developed research interests in trustworthy and secure AI, including prompt injection, adversarial retrieval, and reliable Retrieval-Augmented Generation. Strong programming and software engineering background with experience as a Teaching Assistant for Artificial Intelligence and Data Structures & Algorithms.

EDUCATION

BS Computer Science

Sep 2022 – Jun 2026

COMSATS University Islamabad, Wah Campus

CGPA: **3.65 / 4.00**

Relevant Coursework: Artificial Intelligence, Information Security, Data Structures & Algorithms, Linear Algebra, Calculus

RESEARCH INTERESTS

Trustworthy AI • AI Security • Large Language Models • Retrieval-Augmented Generation • Adversarial Machine Learning • Robust Machine Learning • Deep Learning • Computer Vision

RESEARCH EXPERIENCE

Multi-Stage Traffic Sign Classification Under Heavy Snow

2025

- **Publication Status:** Manuscript in preparation for submission to *MDPI Sensors*.
- Proposed a Task-Aware Multi-Stage Framework for Traffic Sign Classification under snow-degraded conditions.
- Improved EfficientNet-B2 classification accuracy from 68.02% to 87.34% through model optimization for improved robustness.
- Developed practical expertise in deep learning, dataset preparation, model evaluation, experimentation, and scientific writing.

EXPERIENCE

Teaching Assistant

Feb 2025 – Jun 2026

Department of Computer Science, COMSATS University Islamabad, Wah Campus

- Served as a Teaching Assistant across four undergraduate Computer Science courses—**Data Structures & Algorithms, Pattern Recognition, Compiler Construction, and Artificial Intelligence**—during Spring 2025, Fall 2025, and Spring 2026.
- Assisted instructors with the preparation, administration, evaluation, and marking of quizzes, assignments, projects, and examinations.
- Conducted student viva examinations and supported assessment coordination, grading, and maintenance of academic records.
- Provided academic support to students by clarifying course concepts, addressing queries, and assisting with assignments and other course-related activities.

Pakistan Telecommunication Authority (PTA) | ICT Intern

Jul 2025 – Sep 2025

- Developed ASP.NET Web API backend services for an E-Diary system, digitizing internal documentation workflows.
- Designed normalized relational database schemas for efficient data storage and retrieval.
- Developed a Flutter-based frontend integrated with REST APIs, delivering a responsive user experience.

HBL Microfinance Bank Ltd. | Software Quality Assurance Intern

Jul 2024 – Sep 2024

- Performed manual functional, regression, and non-functional testing for web and Android banking applications.
- Reported, tracked, and verified software defects using standard Agile QA procedures.
- Automated repetitive testing tasks utilizing VBA and Excel, improving overall testing efficiency.

PROJECTS

AI Email Copilot — RAG-Based LLM Application

Aug 2026

- Developed an AI-powered Gmail assistant using Python, FastAPI, React, TypeScript, PostgreSQL, and pgvector for intelligent email retrieval, analysis, and response generation.

- Implemented a Retrieval-Augmented Generation (RAG) pipeline using semantic search and **all-MiniLM-L6-v2** embeddings to retrieve relevant historical email context for context-aware responses.
- Integrated Gmail API and OAuth 2.0 for secure email synchronization and multi-account access, with APScheduler-based background processing.
- Built an AI inbox chat interface capable of answering questions using retrieved email sources and generating context-aware email drafts using LLM inference.

OpenHouse & Job Fair Portal (Final Year Project)

Sep 2025 – Jun 2026

- Designed a full-stack recruitment platform using React, Flutter, ASP.NET Core 8 Web API, PostgreSQL, and SignalR.
- Implemented a conflict-free interview scheduling algorithm capable of efficiently allocating hundreds of slots.
- Beta-tested the platform during the Spring 2026 Job Fair, supporting 150+ students and 20+ recruiting companies.
- Deployed the application on AWS EC2 using Docker containers, NGINX reverse proxy, and CI/CD practices.

Event Reminder App

Apr 2025 – May 2025

- Developed a cross-platform mobile application utilizing Flutter for event scheduling and reminders.
- Integrated Firebase Authentication and Cloud Firestore for secure access and real-time data synchronization.

TECHNICAL SKILLS

AI & ML: Machine Learning, Deep Learning, Computer Vision, Large Language Models, Retrieval-Augmented Generation

Programming: Python, C++, C#, JavaScript/TypeScript, PHP, Dart, SQL, VBA

Frameworks: FastAPI, ASP.NET Core 8, React.js, Flutter, Laravel, SignalR

AI/ML Tools: pgvector, Semantic Search, YOLO

Cloud & DevOps: AWS EC2, Docker, Kubernetes, Jenkins, Git, GitHub Actions, Linux, NGINX

Databases: PostgreSQL, MySQL, Oracle Database, Firebase Firestore

HONORS, LEADERSHIP & CERTIFICATIONS

- **Merit Certificate (2025):** Awarded for achieving a perfect 4.00/4.00 SGPA in the 7th semester.
- **Google Cybersecurity Professional Certificate :** Completed an 8-course specialization covering Linux, SQL, networking, security fundamentals, and cybersecurity practices.
- **IELTS Academic:** Overall Band Score **7.0**
- **Student Startup Business Society (SSBS):** Progressed from Operations Team Member to Head of Operations and Advisor, leading recruitment, event planning, and team coordination (Sep 2024 – Jun 2026).
- **Event Organizer & Competitor:** Coordinated university-wide Student Week; participated in VisioSpark Mobile App Development Competition (Nov 2025).

REFERENCES

Dr. Khalid Iqbal

Tenured Professor

Department of Computer Science

COMSATS University Islamabad, Wah Campus

Research Advisor and DAA Instructor

Email: khalidiqbal@ciitwah.edu.pk

Dr. Muhammad Bilal

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Dr. Muhammad Sharif

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Ms. Marwa Khanam

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Ms. Tooba Tehreem

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